House Price Prediction Using Machine Learning

Project Report

Student Name

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Project Title

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1. Introduction

House price prediction is one of the most common regression problems in machine learning. The objective of this project is to build a machine learning model capable of predicting the selling price of houses based on various features such as the overall quality, living area, number of rooms, garage capacity, basement area, and many other property characteristics.

Accurate house price prediction can assist buyers, sellers, real estate agencies, and financial institutions in making informed decisions. This project demonstrates a complete machine learning workflow including data preprocessing, exploratory data analysis, feature engineering, model building, and model evaluation.

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2. Objective

The main objectives of this project are:

- To analyze the house price dataset.
- To clean and preprocess the data.
- To perform Exploratory Data Analysis (EDA).
- To engineer useful features.
- To train multiple machine learning models.
- To compare the performance of different models.
- To identify the best-performing model for predicting house prices.

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3. Dataset Description

The dataset used in this project is the House Prices dataset.

Dataset Information

- Total Training Records: 1460
- Total Features: 81
- Target Variable: SalePrice

The dataset contains numerical and categorical features describing different characteristics of residential houses.

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4. Data Preprocessing

The following preprocessing steps were performed:

- Loaded the dataset.
- Checked dataset dimensions and data types.
- Identified missing values.
- Removed columns containing more than 50% missing values.
- Filled numerical missing values using the median.
- Filled categorical missing values using the mode.
- Checked duplicate records.
- Verified that the cleaned dataset contained no missing values.

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5. Exploratory Data Analysis (EDA)

EDA was performed to understand the relationships between features and the target variable.

5.1 Distribution of SalePrice

This graph shows how house prices are distributed across the dataset.

5.2 Correlation Heatmap

The heatmap illustrates the correlation between numerical variables and helps identify the most influential features.

5.3 Scatter Plot: GrLivArea vs SalePrice

This graph shows the relationship between above-ground living area and house prices.

5.4 Scatter Plot: OverallQual vs SalePrice

This graph illustrates how the overall quality of a house affects its selling price.

5.5 Box Plot of SalePrice

The box plot identifies the spread of house prices and potential outliers.

5.6 Histogram of Numerical Features

Histograms were used to understand the distribution of numerical variables.

5.7 Pair Plot

The pair plot visualizes relationships among selected important numerical features.

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6. Feature Engineering

Several new features were created to improve model performance.

The engineered features include:

- TotalBathrooms
- TotalPorchSF
- HouseAge
- TotalArea

These new features helped provide more meaningful information to the machine learning models.

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7. Data Encoding

Categorical variables were converted into numerical format using One-Hot Encoding so that machine learning algorithms could process them efficiently.

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8. Machine Learning Models

The following regression algorithms were trained:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

The dataset was divided into training and testing sets using an 80:20 ratio.

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9. Model Evaluation

The models were evaluated using the following metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Model Performance

Model	MAE	RMSE	R^2 Score
Linear Regression	20,466.39	52,349.37	0.6427
Decision Tree Regressor	27,946.90	44,212.40	0.7452
Random Forest Regressor	18,252.84	30,104.44	0.8818

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10. Model Comparison

The graph below compares the performance of all three machine learning models.

[Insert Graph 8: Model Comparison Graph Here]

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11. Feature Importance

Random Forest provides feature importance scores that indicate which features contributed most to the prediction.

[Insert Graph 9: Feature Importance Graph Here]

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12. Results

Among the three regression algorithms, the Random Forest Regressor achieved the best performance.

- Best Model: Random Forest Regressor
- MAE: 18,252.84
- RMSE: 30,104.44
- R^2 Score: 0.8818

The Random Forest model outperformed the other models by achieving the highest prediction accuracy and the lowest prediction error.

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13. Conclusion

This project successfully developed a machine learning model for predicting house prices using the House Prices dataset. The project followed a complete machine learning workflow, including data preprocessing, exploratory data analysis, feature engineering, model training, and model evaluation.

Three regression models were developed and compared. Based on the evaluation metrics, the Random Forest Regressor achieved the best performance with an R^2 Score of **0.8818**, an MAE of **18,252.84**, and an RMSE of **30,104.44**. These results indicate that the model can predict house prices with high accuracy for this dataset.

This project demonstrates practical machine learning skills and provides a strong foundation for solving real-world regression problems in domains such as real estate, finance, and business analytics.

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14. Future Scope

The project can be further improved by:

- Applying Hyperparameter Tuning.
- Testing additional regression algorithms such as XGBoost or Gradient Boosting.
- Deploying the model using Flask or Streamlit.
- Building an interactive web application for house price prediction.

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15. References

1. Kaggle – House Prices: Advanced Regression Techniques Dataset

2. Scikit-learn Documentation
3. Pandas Documentation
4. NumPy Documentation
5. Matplotlib Documentation
6. Seaborn Documentation